

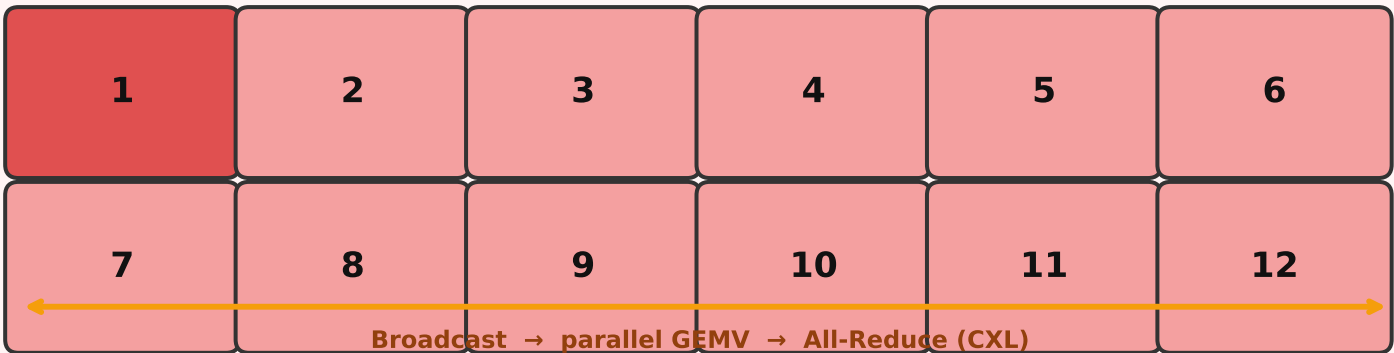
DDA Split: Pool L (TP=12, PP=1) + Pool T (TP=1, PP=80)

Llama2-70B | 80 Decoder Layers | 32 CXL Devices split: N_L=12 + N_T=20
device split

Pool L — Latency Pool

N_L = 12 devices | TP = 12 | PP = 1

Interactive Request



Stage 1: Each device holds 1/12 of FC weights (PP=1 — all 80 layers in one pass)

TTFT = 3.363 s ✓ Meets 5s SLO

Pool L tput = 101.6 tok/s | 1 token in-flight at a time

Tokens served one at a time (no pipeline)

Pool T — Throughput Pool

N_T = 20 devices | TP = 1 | PP = 80

Batch Request (many tokens)

Dev 1	Layers 1-4	← Token 1
Dev 2	Layers 5-8	
Dev 3	Layers 9-12	
Dev 4	Layers 13-16	
Dev 5	Layers 17-20	
Dev 6	Layers 21-24	← Token 2
Dev 7	Layers 25-28	
Dev 8	Layers 29-32	
Dev 9	Layers 33-36	
Dev 10	Layers 37-40	
Dev 11	Layers 41-44	← Token 3
Dev 12	Layers 45-48	
Dev 13	Layers 49-52	
Dev 14	Layers 53-56	
Dev 15	Layers 57-60	
Dev 16	Layers 61-64	← Token 4
Dev 17	Layers 65-68	
Dev 18	Layers 69-72	
Dev 19	Layers 73-76	
Dev 20	Layers 77-80	

Up to 20 tokens in pipeline simultaneously

Pool T tput = 955.8 tok/s

TTFT >> 5s | batch / offline only | 1 output/device-step at steady state

DDA System Throughput = 101.6 + 955.8 = 1057 tok/s vs Static (PP=2, TP=16): 216 tok/s → 4.9× gain